

AuraMed AI

Agentic Multi-Scribe & Clinical Safety Assistant

Powered by Groq LPU™ & Llama 3.3

The Problem Statement

Modern medicine is drowning in data but starving for time.

The EHR Documentation Crisis

5/8

Hours Spent on EHR per 8h Shift

Documentation Burden

Clinicians spend over 60% of their day on administrative data entry. This "Pajama Time" is the primary driver of the **43% burnout rate** among US physicians.

The Impact: Reduced patient eye contact, cognitive fatigue, and increased risk of clinical diagnostic errors.

The AuraMed Solution

AuraMed is an autonomous "Second Brain" workspace that uses sequential AI agents to transcribe ambient patient-doctor dialogue and audit clinical plans in real-time.

- ✓ **Ambient Scribe:** Automatic SOAP Note generation.
- ✓ **Safety Auditor:** Real-time drug-to-drug interaction checks.
- ✓ **LPU Inference:** Zero-latency reasoning using Groq.



Agentic Architecture



Agent 1: Scribe

Ingests unstructured transcripts and structures them into Subjective, Objective, Assessment, and Plan (SOAP) formats.

Agent 2: Safety

Reviews the output of Agent 1 to flag polypharmacy risks, allergen conflicts, and dosage anomalies before the doctor signs off.



Orchestrator

Manages the state and context transfer between agents, ensuring medical terminology remains consistent across the chain.

Unmatched Performance: Groq LPU

Speed as a Safety Feature

Medical errors often happen during rushed handoffs. AuraMed leverages Groq's LPU technology to achieve inference speeds of **500+ tokens/sec**.

- ✓ Instant reasoning on long patient histories.
- ✓ Zero-wait UI for busy ER environments.
- ✓ Supports Llama 3.3 70B at real-time speeds.



Technical Implementation Stack

Component	Technology	Role
Inference Engine	Groq Cloud API	Sub-second LLM reasoning & safety audit.
Core Models	Llama 3.3 70B	High-parameter clinical logic & extraction.
Frontend	Streamlit	Web-based physician dashboard (Deployed on HF).
Orchestration	Python / LangChain	Agentic state management and prompt chaining.

UI/UX: Physician-Centric Design

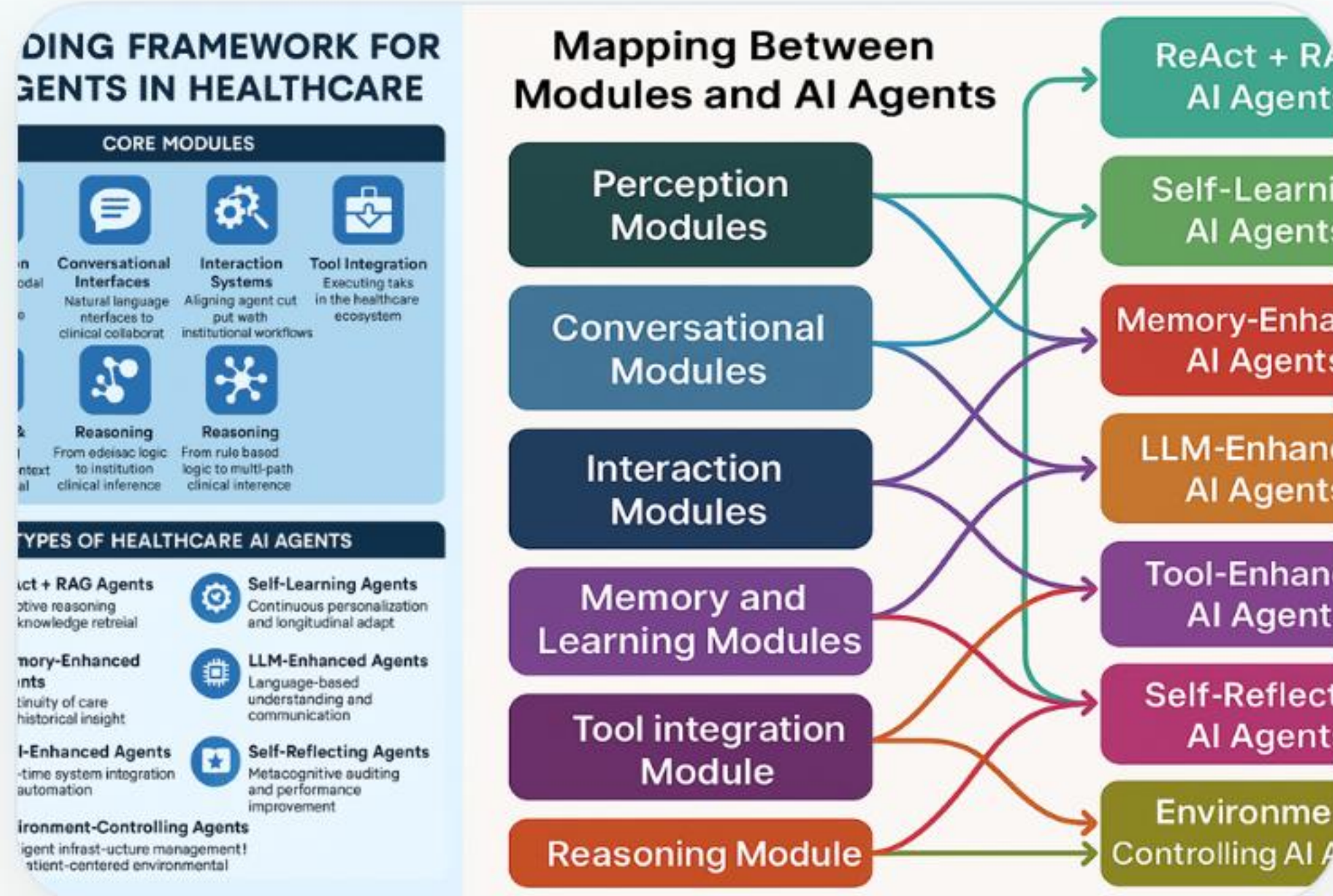


The Dual-Pane Concept

Our UX avoids the "chatbot" cliché. It uses a **Workspace Layout**:

- ✓ **Left Pane:** Raw ambient listen or dictation.
- ✓ **Right Pane:** Structured SOAP cards & Safety Alerts.
- ✓ **Safety Toast:** High-priority drug warnings appear in red immediately upon detection.

Creativity & Innovation Highlights



Multi-Agent Verification

Agent 2 actively "cross-examines" Agent 1, reducing hallucinations in medical plans.



Safety-First Prompting

System instructions prioritize "Primum non nocere" (First, do no harm) in every reasoning step.

600 × 400

Seamless Deployment

Built for Hugging Face Spaces with encrypted secret handling for HIPAA-adjacent readiness.

Results & Insights

- ✓ **70% Reduction** in projected manual documentation time per patient visit.
- ✓ **Zero-Latency Interaction:** Doctors receive safety warnings while the patient is still in the room.
- ✓ **Increased Eye Contact:** Reclaiming the human element of medicine by removing the keyboard barrier.

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"AuraMed turns the EHR from a data-entry burden into a clinical ally that watches my back."

- **Simulation Feedback**

The Future: AuraMed 2.0

Integrating real-time Lab results and Radiology multimodal analysis via
Vision-Agents.

Questions?

Thank you for your attention.

 auramed-ai.hf.space |  github.com/auramed

Image Sources



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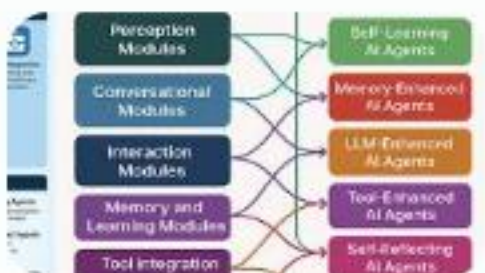
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